

Week 5 : RTT relations $\rightarrow$ Bethe Ansatz, SoV

Week 3 : Particle motion on $AdS_5 \times S^5$

This week : Classical integrable systems

Sigma model on $\mathbb{R} \times S^3 \subset AdS_5 \times S^5$

($\leftrightarrow$ $SU(2)$ sector in $\mathcal{N}=4$ SYM)

Pohlmeyer reduction and giant magnons

Classical spectral curve

§ Classical Integrable System

EoM in classical mechanics $\rightarrow$ nonlinear PDE

$$\dot{q}_i = f_i(p, q), \quad \dot{p}_i = g_i(p, q)$$

SoV (if possible) $\rightarrow$ Action-angle variables

$$\dot{I}_i = 0, \quad \theta_i = \omega_i t$$

$\rightarrow$ Liouville integrability, but how to find it?

A: No axiomatic "proof," many working examples

Idea: express eom as a linear system &
find one-parameter generalization

1) Lax pair

$$\frac{d}{dt} L(\lambda) = [L(\lambda), M(\lambda)]$$

$\lambda \in \mathbb{C}$: spectral parameter

$L(\lambda), M(\lambda)$: matrix or differential op.

$$\frac{d}{dt} \text{tr} L^n = 0 \Rightarrow \text{tr} L^n = \sum_n \lambda_n Q^n$$

2) Zero curvature equation

(Lax connection, Zakharov-Shabat eq.)

$$[\partial_t - L_t(\lambda), \partial_x - L_x(\lambda)] = 0$$

$$\underbrace{\hspace{10em}}_{= M(\lambda) \text{ of Lax pair}}$$

compatibility condition of auxiliary linear problem

$$(\partial_\alpha - L_\alpha(\lambda)) \psi(t, x) = 0, \quad \alpha = t, x$$

Consider Wilson line (monodromy matrix)

$$W(b, a) = \underline{P \exp \int_a^b dx L_x(t, x, \lambda)}$$

path ordered exponential

$$\partial_a W = -W L_x(t, a), \quad \partial_b W = L_x(t, b) W$$

$$\Rightarrow \psi'(x) = W(x, y) \psi(y) \quad \text{satisfies}$$

$$(\partial_x - L_x) \psi' = 0$$

$W(x, y)$ is a gauge transformation

Consider time derivative

$$\partial_t W = \int_a^b ds \quad W(b,s) \quad \underline{\partial_t L_x(t, x=s)} \quad W(s,a)$$

$$= \partial_x L_t + [L_t, L_x]$$

This is total derivative:

$$\partial_s (W(b,s) L_t(s) W(s,a))$$

$$= -W L_x L_t W + W \partial_x L_t W + W L_t L_x W$$

$$= W (\partial_x L_t + [L_t, L_x]) W$$



Thus

$$\partial_t W(b,a) = L_x(x=b) W(b,a) - W(b,a) L_x(x=a)$$

Wilson loop (transfer matrix)

$$T_C(\lambda) = \text{tr} \left(P \exp \left[\oint_C dx L_x(t,x,\lambda) \right] \right)$$

generates the conserved charges,

$$\dot{T}_C = 0, \quad T_C(\lambda) = \sum_n \lambda_n Q^n$$

$$\text{Zero-curvature} \Rightarrow T_C(\lambda) = T_{C+\delta C}(\lambda)$$

as long as connection is smooth.

Examples

1) KdV equation,
$$\begin{cases} L = -\partial_x^2 + u \\ M = 4\partial_x^3 - 6u\partial_x - 3u_x \end{cases}$$

$$\partial_t L = [L, M] \Rightarrow u_t + 6u u_x + u_{xxx} = 0$$

$$L\psi = \lambda\psi : \text{"Schrödinger equation"}$$

$M = M_2$ defines "energy"; $\exists$ higher charges M_n

2) sine-Gordon

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi)^2 + \frac{m^2}{\beta^2} (\cos(\beta\phi) - 1)$$

com: $0 = (-\partial_t^2 + \partial_x^2) \phi - \frac{m^2}{\beta} \sin(\beta\phi)$

rescale $(t, x, \phi) \longrightarrow m = \beta = 1$

Lax connection:

$$L_t = \frac{1}{2} \begin{pmatrix} -i\lambda & \phi_x \\ \phi_x & i\lambda \end{pmatrix}, \quad L_x = \frac{1}{2} \begin{pmatrix} -i/\lambda & \sin \phi \\ \sin \phi & i/\lambda \end{pmatrix}$$

Finding one-soliton solution of SG:

$$T = \frac{t+x}{2}, \quad X = \frac{t-x}{2}$$

$$\partial_t^2 \phi - \partial_x^2 \phi + \sin \phi = 0 \Rightarrow \partial_T \partial_X \phi + \sin \phi = 0$$

Bäcklund transform: ϕ_0 is sol $\Leftrightarrow \phi_1$ is sol

$$\begin{cases} \partial_T (\phi_1 - \phi_0) = 2a \sin\left(\frac{\phi_1 + \phi_0}{2}\right) \\ \partial_X (\phi_1 - \phi_0) = -\frac{2}{a} \sin\left(\frac{\phi_1 - \phi_0}{2}\right) \end{cases} \quad (\forall a \in \mathbb{C})$$

$\Rightarrow$ linear in ∂_T, ∂_X

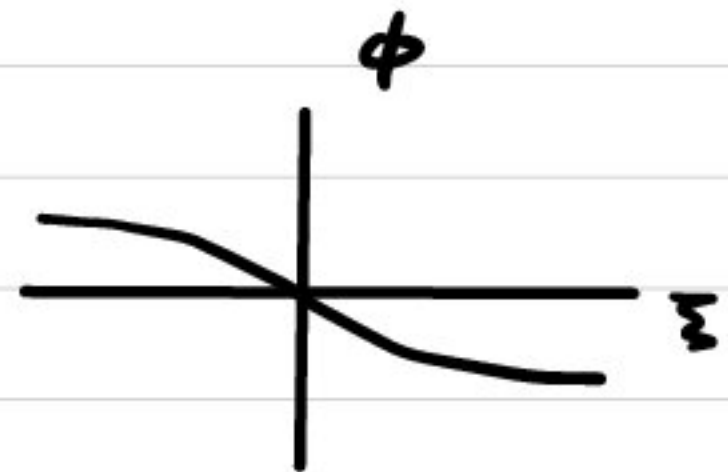
Substitute $\phi_0 = 0$

$$\frac{1}{2a} \frac{\partial}{\partial T} \phi_1 = \sin \frac{\phi_1}{2}, \quad \frac{a}{2} \frac{\partial}{\partial X} \phi_1 = -\sin \frac{\phi_1}{2}$$

$$\xi \equiv \frac{X}{a} - aT, \quad \frac{1}{2} \frac{\partial}{\partial \xi} \phi_1 = -\sin \frac{\phi_1}{2}$$

using $\int d \log \tan \frac{\phi}{4} = \int \frac{1}{2} \frac{d\phi}{\sin \frac{\phi}{2}} = -\xi$

$$\phi_1 = 4 \arctan e^{-\xi}$$



In the original (x, t) coordinate

$$-\xi = \frac{x - vt}{\sqrt{1 - v^2}}, \quad a = \frac{1 - v}{\sqrt{1 - v^2}}$$

velocity $\leftrightarrow$ spectral param. $\lambda = a$

Energy :

$$E = \int \frac{1}{2} (\partial_t \phi)^2 + \frac{1}{2} (\partial_x \phi)^2 + (1 - \cos \phi)$$

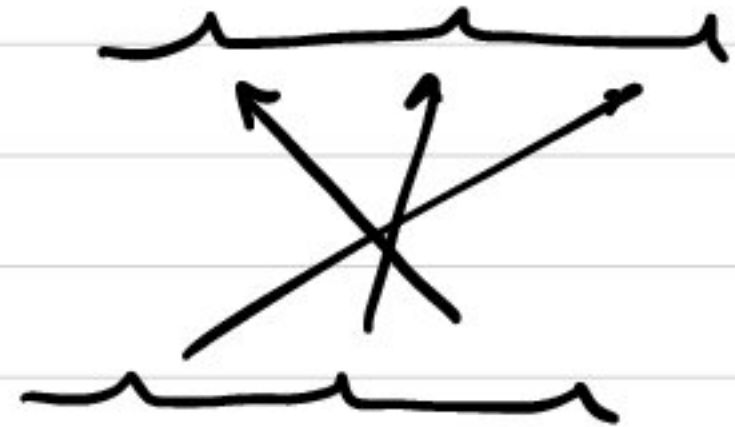
$$= \frac{4}{\sqrt{1 - v^2}} \int d\xi \frac{1}{\cosh^2 \xi} = \frac{8}{\sqrt{1 - v^2}}$$

energy concentrates at $\xi = 0$

Note

- Energy of N solitons

$$E = \sum_{i=1}^N \frac{8}{\sqrt{1-v_i^2}}$$



- Complicated to find the profile of N -solitons

(dressing method is easier than Bäcklund)

{ Classical sigma model on $\mathbb{R}_t \times S^3$ [Dorey, Vicedo
0601194]

(nonlinear) sigma model:

$$S = \frac{R^2}{4\pi\alpha'} \int_M d^2\sigma \sqrt{-\gamma} \gamma^{ab} G_{MN} \partial_a X^M \partial_b X^N$$

Sigma model on $M \simeq G/H$ is classically integrable

If $M \simeq \mathbb{R} \times S^3 \simeq \mathbb{R} \times SU(2)$

$$G \partial_a X \partial_b X = \text{tr}[(g^{-1} \partial_a g)(g^{-1} \partial_b g)], \quad g \in SU(2)$$

"principal chiral model (PCM)"

Parameterization of $g \in \text{SU}(2)$

$$g = \begin{pmatrix} \chi_1 + i\chi_2 & \chi_3 + i\chi_4 \\ -\chi_3 + i\chi_4 & \chi_1 - i\chi_2 \end{pmatrix} = \begin{pmatrix} z_1 & z_2 \\ -\bar{z}_2 & \bar{z}_1 \end{pmatrix}$$

with $1 = \sum_{i=0}^4 |\chi_i|^2 = \sum_{\alpha=1,2} |z_\alpha|^2$

OR

$$g = e^{i\phi\sigma_3} e^{i\theta\sigma_2} e^{i\psi\sigma_3} = \begin{pmatrix} \cos\theta e^{i(\phi+\psi)} & \sin\theta e^{i(\phi-\psi)} \\ -\sin\theta e^{-i(\phi-\psi)} & \cos\theta e^{-i(\phi+\psi)} \end{pmatrix}$$

Use

- $X_0 = k\tau$ for time in $\mathbb{R}_t \times S^3$ (Euclidean)

- $\sqrt{\lambda} \equiv R^2/\alpha'$ is 't Hooft coupling

- $\hat{j}_a = g^{-1} \partial_a g$, $su(2)$ current (Maurer-Cartan)

$$* \hat{j}_a = \epsilon^{ab} \hat{j}_b \quad \text{with } a, b = 1, 2$$

- choose conformal gauge in Euclidean w.s.

$$S = - \frac{\sqrt{\lambda}}{4\pi} \int \left[dX_0 \wedge * dX_0 + \frac{1}{2} \text{tr} (\hat{j} \wedge * \hat{j}) \right]$$

Flatness condition:

$$\text{from } \hat{j} = g^{-1} dg, \quad dj = dg^{-1} \wedge dg = -\hat{j} \wedge j$$

$$\text{EoM: from } g \rightarrow (1+\epsilon)g, \quad g^{-1} \rightarrow g^{-1}(1-\epsilon)$$

$$j \rightarrow \hat{j} + d\epsilon$$

$$\text{integration by parts} \leadsto d \star \hat{j} = 0$$

$$\text{Virasoro condition: } \text{tr}(j_+)^2 = \text{tr}(j_-)^2 = -2k^2$$

$$j_{\pm} = \hat{j}_0 \pm j_1$$

Spacetime energy : $E = \frac{\sqrt{\lambda}}{2\pi} \oint d\sigma \partial_t = \sqrt{\lambda} K$

$SU(2)_L \times SU(2)_R$ symmetry ;

$$g \rightarrow U_L g U_R, \quad U = \begin{pmatrix} e^{i\alpha} & 0 \\ 0 & e^{-i\alpha} \end{pmatrix}, \quad \begin{cases} z_1 \rightarrow e^{i\alpha_R + i\alpha_L} z_1 \\ z_2 \rightarrow e^{i\alpha_R - i\alpha_L} z_2 \end{cases}$$

Noether charge :

$$J_1 + J_2 = \frac{\sqrt{\lambda}}{4\pi i} \oint \text{tr} (g^{-1} \partial_0 g \cdot \sigma_3)$$

$$J_1 - J_2 = \frac{\sqrt{\lambda}}{4\pi i} \oint \text{tr} (\partial_0 g g^{-1} \cdot \sigma_3)$$

Lax connection:

$$J(x) \equiv \frac{\hat{j} - x * j}{1 - x^2}$$

flatness:

$$\Rightarrow dJ(x) - J(x) \wedge J(x) = 0$$

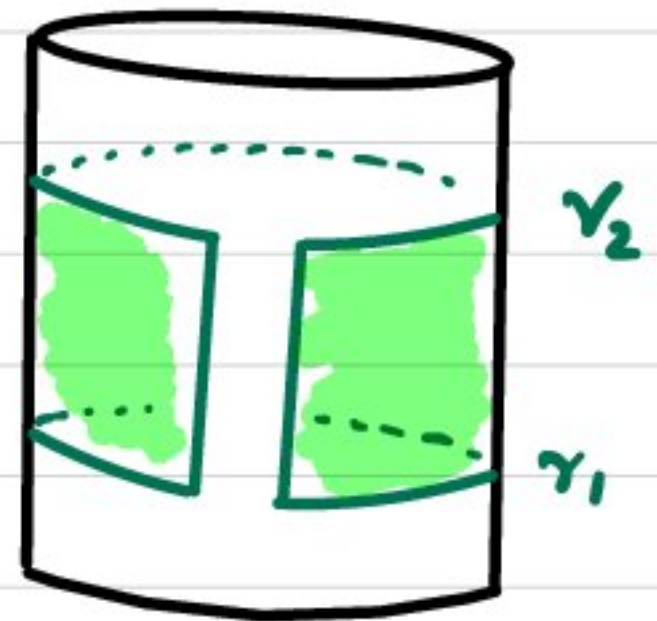
$$J_+(\tau, \sigma, x) = \frac{1}{2} \left(\frac{\hat{j}_+}{1-x} - \frac{j_-}{1+x} \right)$$

Monodromy:

$$\Omega(x) = P \exp \left(\oint_{\gamma} d\sigma J_+(\tau, \sigma, x) \right)$$

$\Omega(x)$ is independent of reference point

(τ, σ) because of flatness condition



$$\int d\sigma d\tau (dJ - J \wedge J) = 0$$

Classification of all solutions of PCM

$\longleftrightarrow$ all possible forms of quasi-momentum $p(x)$

We skip construction of string profile $X^M(\tau, \sigma)$

(but not easy to compute $p(x)$ from a given $X^M(\tau, \sigma)$)

Profile $\leadsto$ Baker - Akhiezer function

(Fay identity on Riemann Θ function + uniqueness)

$\Omega = P \exp \left(\oint J_+ \right)$ is 2×2 matrix

Matrix elements are meromorphic in x (except $x = \pm 1$)

Diagonalize $\Omega \rightarrow p(x)$: quasi-momentum

$$U \Omega U^{-1} = \text{diag} \left(e^{ip(x)}, e^{-ip(x)} \right)$$

$p(x)$ has $\left\{ \begin{array}{l} \text{simple poles at } x = \pm 1 \\ \text{square-root branch cuts, } p \leftrightarrow -p \end{array} \right.$

§ Pohlmeyer reduction and giant magnons

Operators in $N=4$ SYM

[Hofman, Maldacena
0604135]

$$\mathcal{O} \sim \sum_{n_1, n_2} e^{ip_1 n_1 + ip_2 n_2} \text{tr} \left(Z Z Z \dots \underset{\hat{n}_1}{W} \dots \underset{\hat{n}_2}{W} \dots Z Z Z \right)$$

W is "excitation" or "magnon"

In classical string theory, we expect

magnon $\longleftrightarrow$ soliton solution in $AdS_5 \times S^5$
"giant magnon"

"Soliton" on

$\mathbb{R} \times S^2$

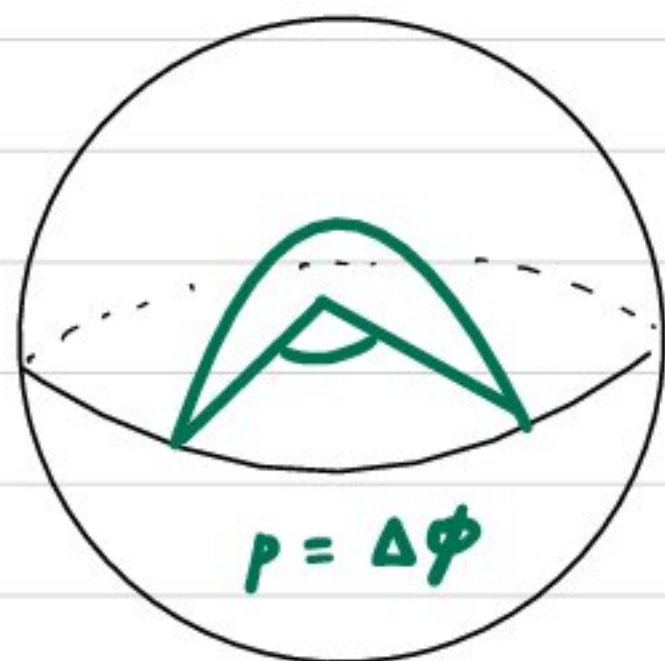
$\xrightarrow{\text{Pohlmeyer reduction}}$

soliton of

Sine-Gordon eq.

$$\partial_+ \partial_- \vec{X} + \cos \phi \vec{X} = 0$$

$$\partial_+ \partial_- \phi + \sin \phi = 0$$



$\xleftarrow{\text{inverse}}$

$$v = \cos \frac{p}{2}$$

p : magnon momentum, $E \sim \frac{\sqrt{\lambda}}{2} \sin \frac{p}{2}$

$$\phi = 4 \arctan e^{\frac{x-vt}{\sqrt{1-v^2}}}$$

v : spectral parameter

Derivation of Pohlmeyer reduction

Write the sigma model action as

$$L \sim \partial_+ \vec{X} \cdot \partial_- \vec{X} + \underbrace{\Lambda (\vec{X} \cdot \vec{X} - 1)}_{\text{Lagrange multiplier}}$$

Constraint: $\vec{X} \cdot \vec{X} = 1 \Rightarrow \partial_+ \vec{X} \cdot \vec{X} = \partial_- \vec{X} \cdot \vec{X} = 0$

Virasoro conditions: $(\partial_+ \vec{X})^2 = (\partial_- \vec{X})^2 = K^2$

from $t = K\tau$, choose $K = 1$

eom: $\partial_+ \partial_- \vec{X} + \Lambda \vec{X} = 0$

multiply $\vec{X}$: $\vec{X} \cdot \partial_+ \partial_- \vec{X} + \Lambda = 0$

differentiate $\vec{X} \cdot \partial_- \vec{X} = 0 \leadsto \Lambda = \partial_+ \vec{X} \cdot \partial_- \vec{X} = \cos \phi$

$$\Rightarrow \partial_+ \partial_- \vec{X} + \cos \phi \vec{X} = 0$$

Main idea :

use $(\vec{X}, \partial_+ \vec{X}, \partial_- \vec{X})$ as a basis of $S^2 \subset \mathbb{R}^3$

derive eom for $\partial_+ \vec{X} \cdot \partial_- \vec{X} = \cos \phi$

Ansatz : $\partial_+^2 \vec{X} = a \vec{X} + b \partial_+ \vec{X} + c \partial_- \vec{X}$

Apply $\vec{X}$; $\vec{X} \cdot \partial_+^2 \vec{X} = \partial_+ (\vec{X} \cdot \partial_+ \vec{X}) - (\partial_+ \vec{X})^2$
 $\leadsto a = -1$

" $\partial_+ \vec{X}$: $\partial_+^2 \vec{X} \cdot \partial_+ \vec{X} = b + c \cos \phi = 0$

" $\partial_- \vec{X}$: $\partial_+^2 \vec{X} \cdot \partial_- \vec{X} = \partial_+ (\partial_+ \vec{X} \cdot \partial_- \vec{X}) + \partial_+ \vec{X} \cdot \vec{X} \cos \phi$

$\partial_+ (\cos \phi) = b \cos \phi + c \leadsto \text{solve } b, c$

Same equations by interchanging $+ \leftrightarrow -$

Result :
$$\partial_+^2 \vec{X} = -\vec{X} + \frac{\cos \phi}{\sin \phi} \partial_+ \vec{X} - \frac{\partial_+ \phi}{\sin \phi} \partial_- \vec{X}$$

$$\partial_-^2 \vec{X} = -\vec{X} + \dots$$

Then :

$$\partial_- \partial_+^2 \vec{X} = \left[-1 - \partial_- \left(\frac{\partial_+ \phi}{\sin \phi} \right) \right] \partial_- \vec{X}$$

$$+ \partial_- \partial_+ (\log \sin \phi) \partial_+ \vec{X} + \partial_+ (\log \sin \phi) \cdot \cos \phi \vec{X}$$

$$- \frac{\partial_+ \phi}{\sin \phi} \underline{\partial_-^2 \vec{X}}$$

substitute the above eq.

compare coeffs of $(\vec{X}, \partial_+ \vec{X}, \partial_- \vec{X})$

(skip details) $\leadsto \partial_+ \partial_- \phi + \sin \phi = 0$

Dyonic giant magnon $\longleftrightarrow$ magnon bound states

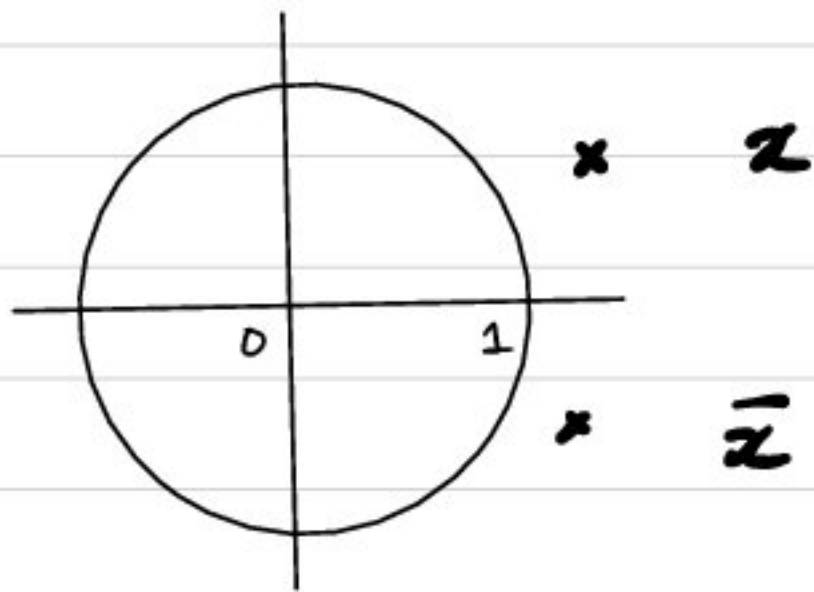
angular momentum

$$J_1 \rightarrow \infty, J_2 = Q$$

$$\left. \begin{array}{c} \times \\ \times \\ \hline \times \\ \times \end{array} \right\} Q$$

$\longleftrightarrow$ Sigma model on $\mathbb{R} \times S^3 \longleftrightarrow$ complex sine Gordon

Classical spectral curve = poles in x -plane



$$\left\{ \begin{array}{l} E - J_1 = \frac{\sqrt{\lambda}}{4\pi} \left(x - \frac{1}{x} - \bar{x} + \frac{1}{\bar{x}} \right) \\ J_2 = \frac{\sqrt{\lambda}}{4\pi} \left(x + \frac{1}{x} - \bar{x} - \frac{1}{\bar{x}} \right) \end{array} \right.$$

Use $\frac{z}{\bar{z}} = e^{ip} \Rightarrow E - J_1 = \sqrt{J_2^2 + \frac{\lambda}{\pi^2} \sin^2 \frac{P}{2}}$

Recall in XXX Bethe Ansatz:

$$\frac{u + i/2}{u - i/2} = e^{ip} \Rightarrow u = \frac{1}{2} \cot \frac{P}{2}$$

$$\Delta = \frac{\lambda}{2\pi^2} \sum_j \frac{1}{4u_j^2 + 1} = \frac{\lambda}{2\pi^2} \sin^2 \frac{P}{2}$$

→ agrees with $E - J_1$ at small λ

← "BPS" relation

§ Finite gap solution

Find various relation for $p(x)$

$$\Omega(x) = P \exp \oint \frac{1}{2} \left(\frac{j_+}{1-x} - \frac{j_-}{1+x} \right)$$

$$\text{tr } j_{\pm} = 0, \quad \text{tr } j_{\pm}^2 = -2k^2$$

$$\Rightarrow \text{diagonalize } j_{\pm} \sim \begin{pmatrix} ik & 0 \\ 0 & -ik \end{pmatrix} \text{ near } x = \pm 1$$

$$\text{Thus } p(x) = -\frac{\pi k}{x \mp 1} + \dots \quad (x \rightarrow \pm 1)$$

Near $x \rightarrow \infty$, we find $SU(2)_R$ global charge

$$\Omega(x) = P \exp \oint \left(-\frac{j_0}{x} + \dots \right) \quad (x \rightarrow \infty)$$

using $\text{tr}(j_0) = 0$, $P \exp \oint \text{tr}(j_0 \sigma_3) = J_1 + J_2$

$$p(x) = -\frac{2\pi}{x\sqrt{\lambda}} (J_1 + J_2) \quad (x \rightarrow \infty)$$

$$= -\frac{2\pi}{x\sqrt{\lambda}} (L - 2J) \quad \begin{cases} L \equiv J_1 - J_2 \\ J = -J_2 \end{cases}$$

Near $x \rightarrow 0$, we find $SU(2)_L$ global charge

$$\partial_a - J_a \sim g^{-1} (\partial_a - x l_a + \dots) g$$

$$l_a = (\partial_a g) g^{-1} \quad \text{gives} \quad J_1 - J_2 = L$$

$$\leadsto p(x) = \underline{2\pi m} + \frac{2\pi x}{\sqrt{\lambda}} L + \dots \quad (x \rightarrow 0)$$

$m \in \mathbb{Z}$, mode number (or winding number)

$p(x)$ is defined modulo $2\pi\mathbb{Z}$, so mode number can be removed from $p(x \rightarrow \infty)$

Classical spectral curve:

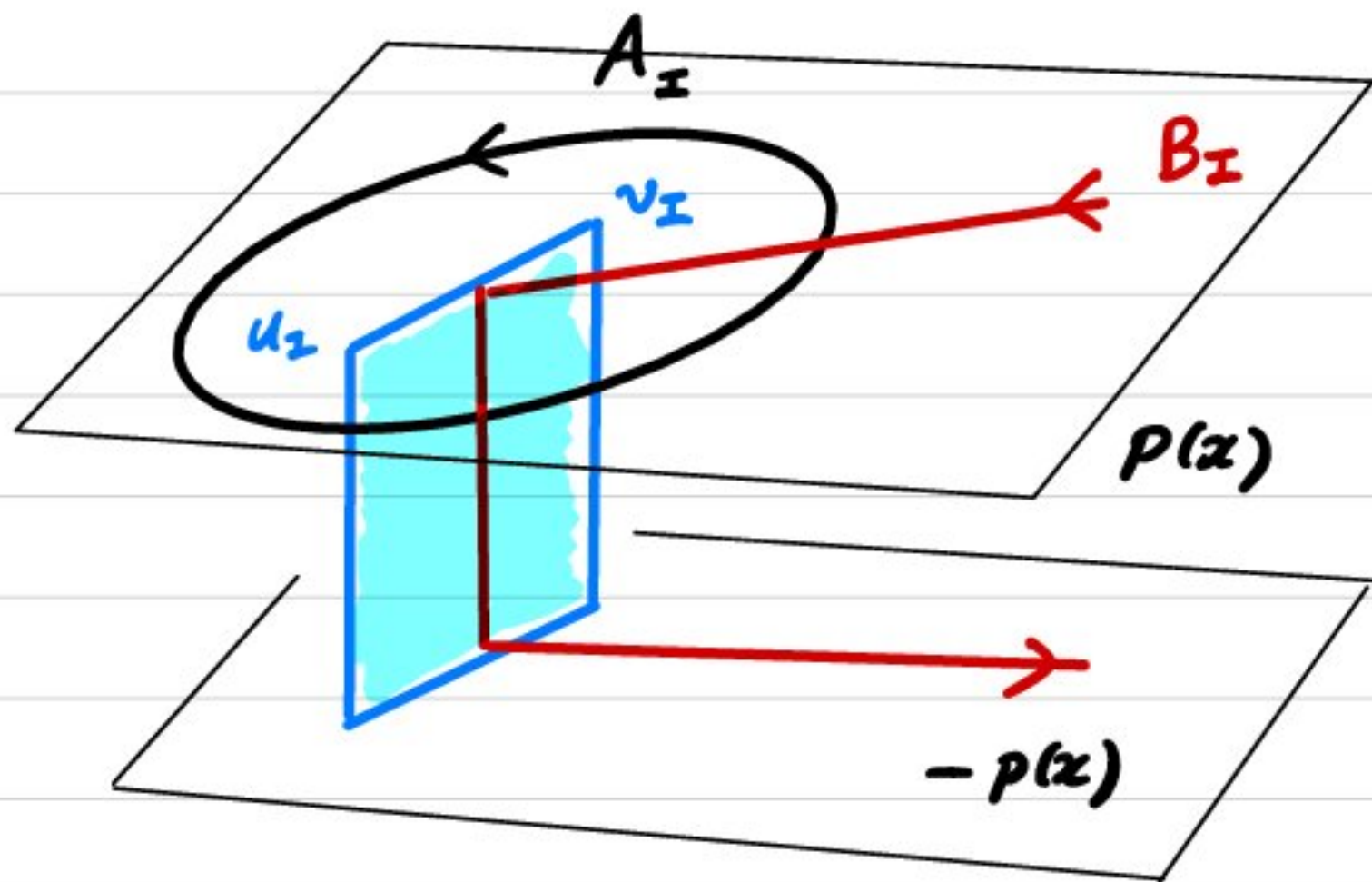
$$\Sigma : \det \begin{pmatrix} y - p'(x) & 0 \\ 0 & y - p'(x) \end{pmatrix} = 0$$

for example, genus- g hyperelliptic curve is

$$y^2 = \prod_{i=1}^{g+1} (x - u_i)(x - v_i)$$

N.B. unlike SW curve, $dp = \frac{dx}{y}$, we fix $p(x)$

from the consistency conditions & analyticity



hyperelliptic curve

has $2g$ cycles

$$\{A_I, B_I\}$$

$$C_I = (u_I, v_I)$$

branch cut

period integrals are quantized

$$\oint_{A_I} dp = 2\pi m_I, \quad \oint_{B_I} dp = 2\pi n_I \quad (m_I, n_I \in \mathbb{Z})$$

Examples of finite-gap solutions $\left[\begin{smallmatrix} \text{KMMZ} \\ 0402207 \end{smallmatrix} \right]$

Two formulations look "similar" in both sides:

Classical spectral curve $\longleftrightarrow$ XXX BAE

String side: $G(z) = p(x) + \frac{\pi \Delta}{\sqrt{\lambda}} \left(\frac{1}{z-1} + \frac{1}{z+1} \right)$

$$p(x) = 2\pi m + \frac{2\pi L}{\sqrt{\lambda}} x + \mathcal{O}(x^2)$$

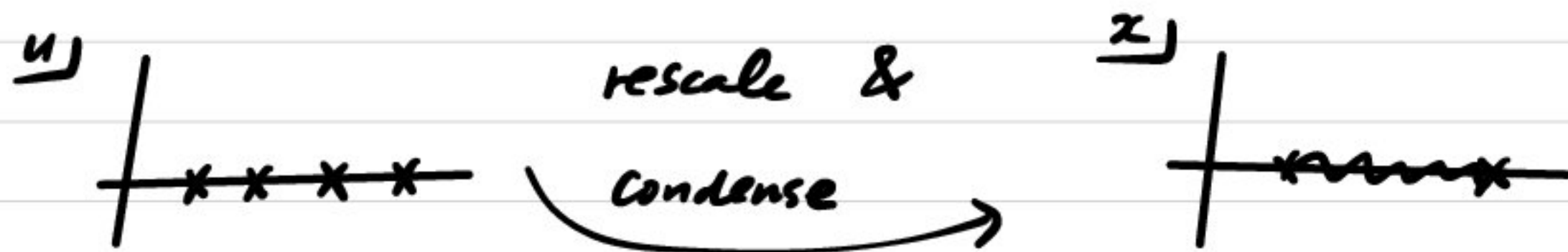
$$\Rightarrow \Delta - L = -\frac{\sqrt{\lambda}}{2\pi} \oint \frac{dx}{2\pi i} \frac{G(z)}{x^2} \quad (\lambda \gg 1)$$

Gauge side: $\gamma = \frac{1}{2\pi^2} \sum_j \frac{1}{4u_j^2 + 1}$

rescale $u_j = L x_j$, $L \rightarrow \infty$, x_j fixed

define resolvent $G(x) = \frac{1}{L} \sum_j \frac{1}{x - x_j} = \int_C \frac{d\zeta \rho(\zeta)}{x - \zeta}$

$\rho(\zeta)$: density of roots



$$\gamma = \frac{\lambda}{8\pi^2 L} \int_C \frac{dx \rho(x)}{x^2} = \frac{-\lambda}{8\pi^2 L} \oint_C \frac{dx}{2\pi i} \frac{G(x)}{x^2} \quad (\lambda \ll 1)$$

$\Rightarrow \Delta-L$ ($\lambda \gg 1$) and γ ($\lambda \ll 1$) are given by

$$\oint \frac{dx}{2\pi i} \frac{G(x)}{x^2}, \quad G(x): \text{resolvent}$$

We can compare other quantities

$$\int dx \rho(x) = \frac{J}{L} : \text{filling fraction (number of roots)}$$

Comment

- Spectral parameter in XXX BAE is u

$$T(u) \quad u = i/2 \leadsto \text{local charge}$$

$$u = \infty \leadsto \text{global charge}$$

- Spectral parameter in classical string is x

$$e^{\pm i p(x)} \quad x = \pm 1 \leadsto \text{energy}$$

$$x = 0, \infty \leadsto \text{global charge}$$

Related by $u = x + \frac{1}{x}$

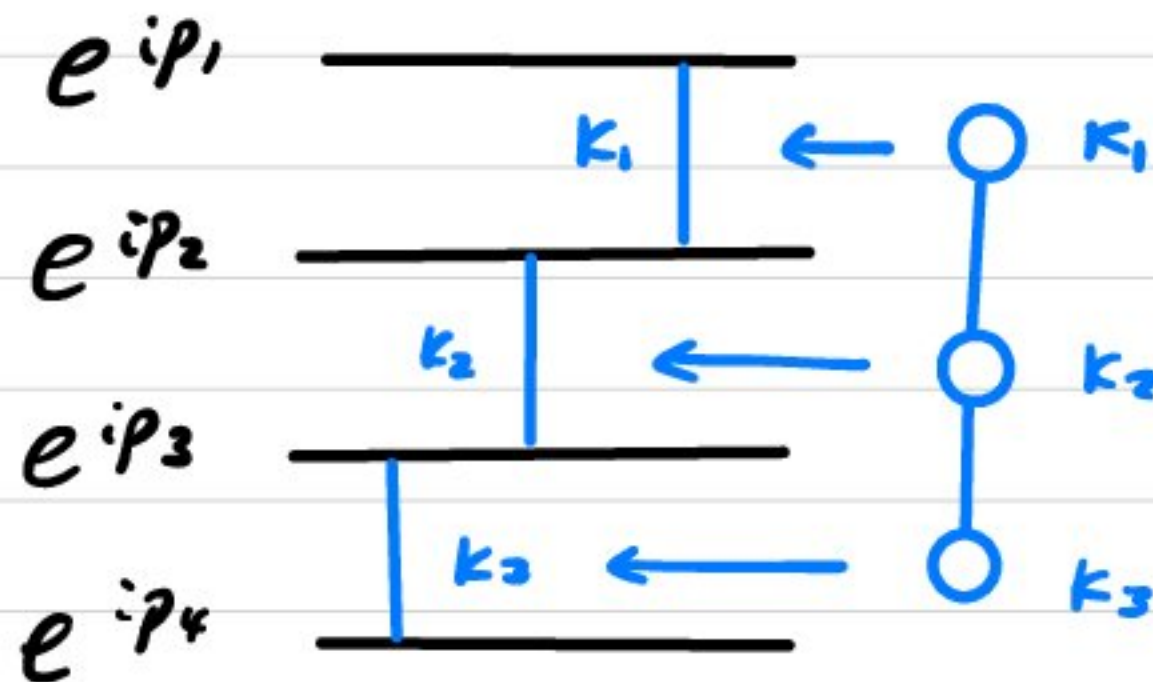
Q: In Bethe Ansatz, only the density of roots is used

Why introduce 2-sheet complex plane in PCM?

Quasi-momentum is the fundamental rep. of $SU(N)$

Bethe root lives in the Dynkin node of $SU(N)$

e.g. $N=4$



"going to another sheet" doesn't exist in XXX